11-06 第2篇

阅读理解（说明文） Why Does Bread Rise? The Small Invisible Friends Inside a Loaf

① If you walk into a bakery on a cold morning, the warm smell will almost catch you by the sleeve. A **golden, risen loaf** of bread sits on the wooden shelf, twice as tall as the small bowl of flour the baker started with. How did that happen? Flour, water and a little salt by themselves cannot grow. Who, or what, pushed the bread upwards?

② The answer is surprisingly tiny, surprisingly alive, and about **5,000 years** old in human kitchens. The workers responsible are called **yeast** (酵母). Yeast is a simple **one-celled living thing** — a fungus. A single yeast cell is about **5 micrometres** wide, far too small to see with the eye. Yet one small packet of dried yeast in the kitchen contains **billions** of these quiet little friends. They have slept in the packet. When we add them to warm water, they wake up.

③ Yeast needs three things to be happy: **warmth, water, and sugar**. Flour already contains a little sugar, and the warm water softens the yeast so it can start to eat. Like all living things, yeast **eats** and **breathes**. It eats the small sugars in the dough. In return, it lets out two things: a gas called **carbon dioxide** (二氧化碳), and a tiny amount of **alcohol**. The alcohol later turns into the warm, slightly toasty smell of good bread; the carbon dioxide, however, is the main actor on today's stage.

④ Here is the clever point: when yeast releases carbon dioxide inside a piece of **wet, sticky dough**, the gas cannot escape to the air. It is trapped inside tiny **air bubbles** in the dough. As more and more gas is made, each bubble grows a little larger, pushing the dough around it outwards. The dough **slowly rises**, like a small grey-white balloon being filled with thousands of little breaths. This is why bakers let the dough rest, sometimes for **one hour, sometimes three**, in a warm corner of the kitchen.

⑤ But a bubble alone is not enough. If you blew into sand, the sand would not hold the shape. The dough holds the shape because of a special substance called **gluten** (面筋). Gluten is a stretchy network of proteins formed when flour and water are mixed and then pushed and folded (that is, **kneaded**) for a few minutes. Gluten acts like a **small invisible net**: it catches the carbon-dioxide bubbles and keeps them inside the dough. The more carefully the dough is kneaded, the stronger the net, and the taller the final loaf can grow.

⑥ Finally the dough goes into the hot oven. Three things happen in the first few minutes at **200 °C**. First, the trapped gas suddenly expands — the dough jumps up one last time, called the "**oven spring**". Second, the yeast, which loves warmth but hates real heat,

dies. Third, the outer surface of the dough dries and browns into a crust. Inside, the gas bubbles are frozen in shape by the baked gluten; outside, a thin golden skin keeps everything in. **A loaf is, in a way, the photograph of the last breath a yeast cell took.**

⑦ Not all bread needs yeast. Some kinds of bread are called "**flat bread**" (for example, pita, chapati, or certain Chinese pancakes). They do not use yeast and do not rise much; they are thin, quick, and still delicious. Some modern bakers also use **sourdough starters** — a slow, old culture of wild yeasts and friendly bacteria grown at home, sometimes fed over many years. Sourdough rises more slowly and tastes a little more complex. A starter can, with good care, **live longer than its baker**, and be passed down from mother to daughter.

⑧ So, the next time you break open a slice of bread and look at the small, uneven holes inside, remember: each hole is a little room once occupied by a living cell, breathing out quiet gas in the warm kitchen of last night. **Bread rises not because we push it, but because we give tiny friends the right place to breathe.** It is a small lesson that applies well beyond the kitchen: in classrooms, in families, in any group of young people. Give a warm, safe place, and things that look flat today may, by tomorrow, quietly rise — golden, surprising, and **twice as tall**.

注释： yeast 酵母；fungus 真菌；carbon dioxide 二氧化碳；gluten 面筋；knead 揉；crust 面包皮

根据短文内容，选择最佳答案。

1. What is yeast, according to Paragraph ②?

A. A kind of salt used in baking.
B. A simple one-celled living thing (a fungus) about 5 micrometres wide, of which a single packet can contain billions.
C. A small insect that lives in flour.
D. A plant with green leaves.

2. What three things does yeast need to be happy?

A. Cold, flour, oil.
B. Warmth, water, and sugar.
C. Salt, pepper, and fire.
D. Sunlight, air, and milk only.

3. Why do the gas bubbles stay inside the dough instead of escaping into the air?

A. Because the dough is too cold.
B. Because of gluten — a stretchy network of proteins formed by kneading — that acts like a small invisible net to catch the carbon-dioxide bubbles.

C. Because the dough is filled with oil.
D. Because the baker pushes the dough.

4. In a hot oven at about 200 °C, which of the following does NOT happen in the first few minutes?

A. The trapped gas suddenly expands, creating the "oven spring".
B. The yeast, which loves warmth but hates real heat, dies.
C. The outer surface of the dough dries and browns into a crust.
D. The yeast swims out of the dough through tiny holes.

5. What is the "small lesson" the writer draws in the last paragraph?

A. Bread is always better than rice.
B. Only old bakers can make good bread.
C. If given a warm, safe place to breathe — whether in a kitchen, a classroom or a family — things that look flat today can, by tomorrow, quietly rise.
D. Students should not bake at home.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "simple one-celled living thing — a fungus ... about 5 micrometres ... billions"
【解析】定义与数量直引。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "warmth, water, and sugar"
【解析】三要素直引。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第⑤段 "gluten ... stretchy network ... small invisible net"
【解析】面筋网络捕气原理。A、C、D 均错。故选 B。

4.
【答案】D
【题型】细节理解题（否定）
【定位】第⑥段三变化
【解析】A、B、C 均在文中；D 与事实相反（酵母被热杀死，不会游出面团）。故选 D。

5.

【答案】C

【题型】主旨大意题

【定位】第⑧段结尾 "Give a warm, safe place ... quietly rise"

【解析】科学延伸到成长主题。A、B、D 均偏离。故选 C。